

Project M2 (Part 2): E-commerce Sentiment Analysis

Automated NLP Segregation of Customer Feedback & Reviews

1. Problem Statement & Objective

Context: ABC E-commerce marketplace receives a massive influx of daily customer feedback. Manual processing of these comments is resource-heavy, expensive, and scales poorly as the business grows.

Objective: Develop an automated AI/NLP model to parse through thousands of comments and accurately segregate them into **Positive** and **Negative** categories. The core focus is to isolate negative feedback rapidly so the service management team can act on it immediately.

2. Model Processing Results

The Python-based Sentiment Automation Script successfully processed the dataset. Below are the execution metrics:

8,005

TOTAL COMMENTS ANALYZED

4,062

POSITIVE FEEDBACKS (50.7%)

3,910

NEGATIVE FEEDBACKS (48.8%)

*Note: ~33 comments were neutralized or blank during processing.

3. Generated Deliverables

File Name	Type	Description
nlp_model.py	Python Code	The automation script that maps raw polarity tags to human-readable Sentiment Categories.
Segregated_Comments.csv	Dataset	The complete database with a newly appended 'Sentiment_Category' column for downstream Tableau/BI reporting.
Critical_Negative_Comments.csv	Actionable List	A filtered database containing ONLY the 3,910 negative comments. Created specifically for the Quality Assurance team to review and resolve.

4. Business Impact & Recommendation

Immediate Action Required:

The model has isolated **3,910 negative comments**. By automating this segregation, we have saved approximately 160 man-hours of manual reading. It is highly recommended that the customer service team integrates `Critical_Negative_Comments.csv` into their daily ticketing system to address these grievances and improve the platform's overall NPS (Net Promoter Score).